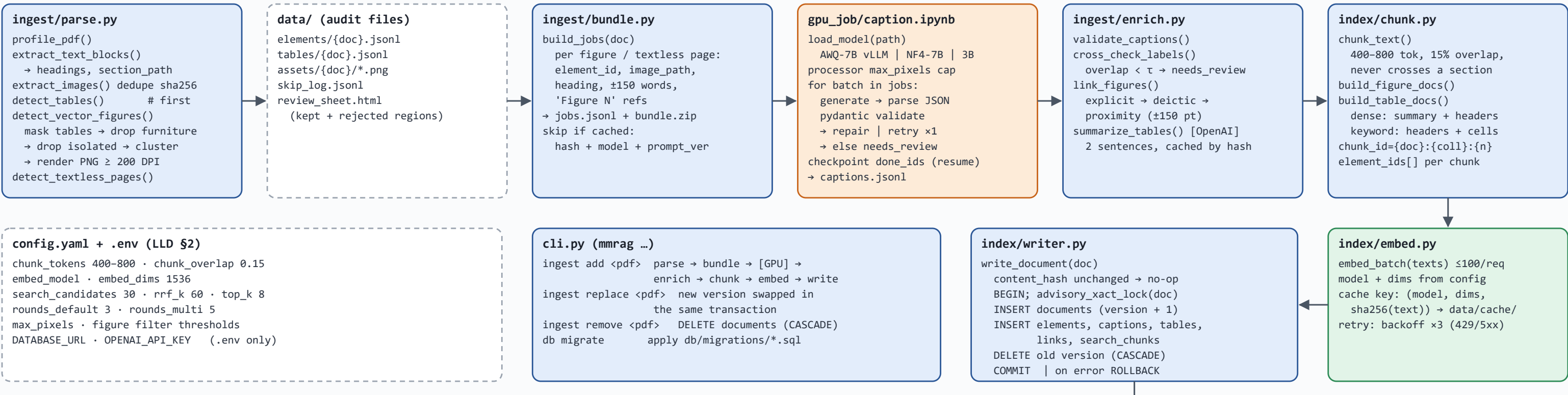


Agentic Multimodal RAG: Low-Level Design

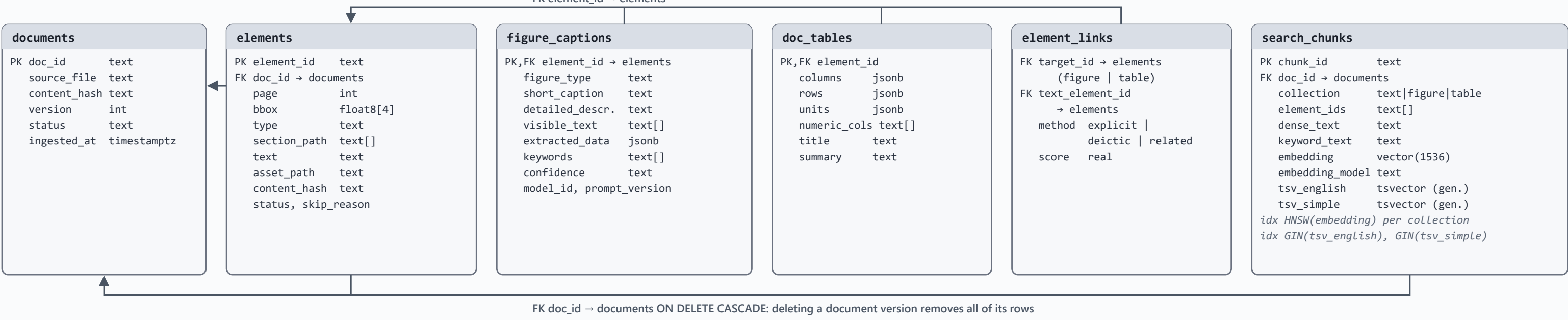
Module-level flow · function responsibilities · PostgreSQL schema · query path. Companion to 05-low-level-design.md (section numbers refer to it).

 Python module (laptop CPU)  GPU job (Kaggle / Colab)  Calls OpenAI API  PostgreSQL table / data  Files on disk / config  UI / answer

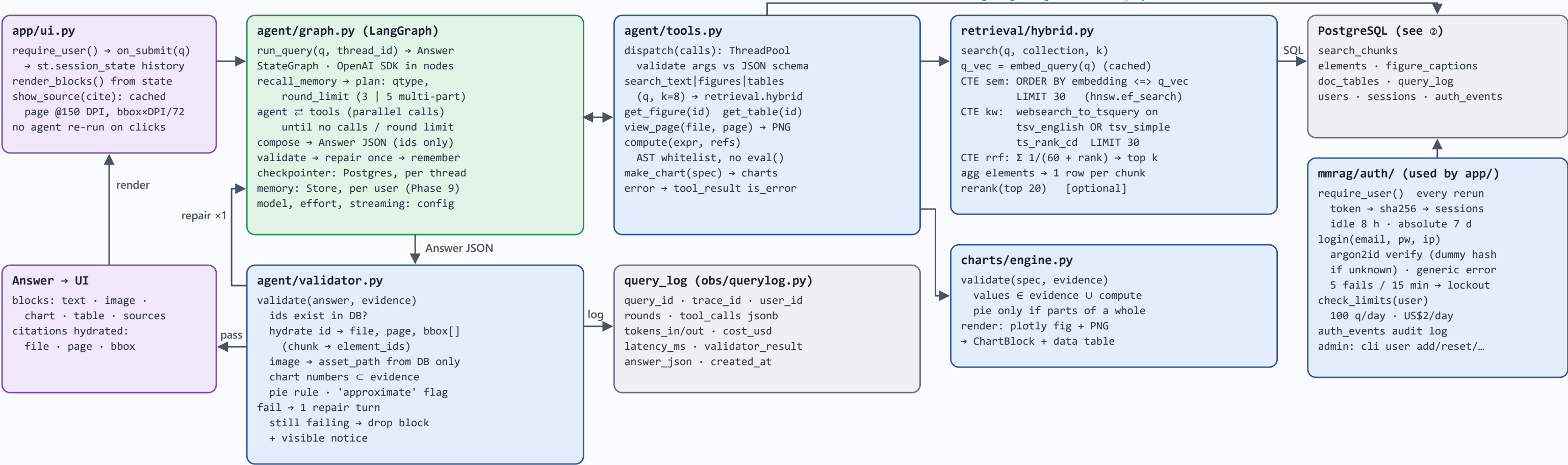
① INGESTION PATH: modules and functions (LLD §3)



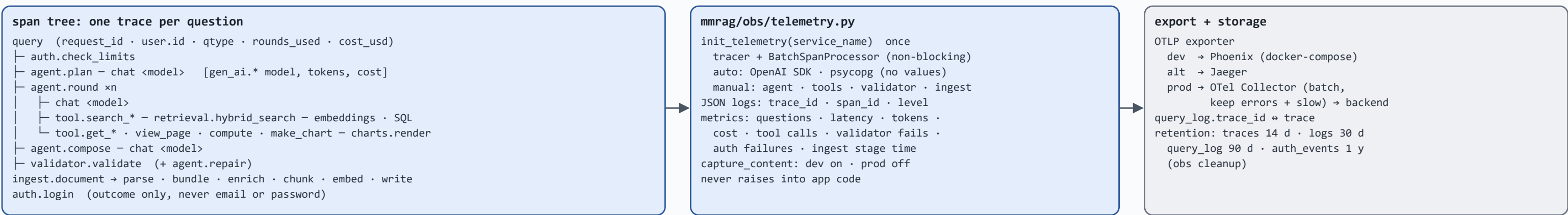
② POSTGRESQL SCHEMA: pgvector + full-text (LLD §4) · dev: pgvector/pgvector:pg16 in a named Docker volume · + users, sessions, auth_events, LangGraph checkpoints + memory store (LLD §4.1)



③ QUERY PATH: modules and functions (LLD §5)



④ OBSERVABILITY: OpenTelemetry traces, logs, metrics (LLD §5.9)



⑤ EVALUATION: RAGAS + custom metrics (LLD §5.10)

